Pregnancy Morbidity & Mortality

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Learning Objectives

- Define pregnancy-related morbidity and mortality using WHO standards
- Describe the impact and recent trends in pregnancy-related mortality in the US
- Recognize disparities in pregnancy-related death
- Identify the factors influencing pregnancy-related morbidity and mortality in the US
- Summarize the work being done to decrease pregnancy-related morbidity and mortality, including continued research and implementation of safety initiatives

We encourage you to. . .

01

Ask questions- please interrupt me often! I love engaging with our participants!

02

Speak from your own experience - share stories that underscore any of the information we share with you today.

03

Maintain a curious, questioning open mind. We invite discussion and welcome your perspective on the information we share today.

04

Take care of yourself and acknowledge your own value in the work you are doing to make a difference in addressing reproductive health.

Language Matters!

LESS INCLUSIVE LANGUAGE	**maternal morbidity and mortality**	**pregnant woman**	**maternal health**
MORE INCLUSIVE LANGUAGE	pregnancy-related morbidity and mortality	pregnant person	reproductive and sexual health

Definitions: WHO

- **Pregnancy-related morbidity:**
  - any health condition attributed to and/or complicating pregnancy, and childbirth that has a negative impact on well-being and/or functioning
- **Pregnancy-related death:**
  - annual number of deaths from any cause related to or aggravated by pregnancy or its management (excluding accidental or incidental causes) during pregnancy and childbirth or within 42 days of termination of pregnancy, irrespective of the duration and site of the pregnancy.
- **Late pregnancy-related death:**
  - death from direct or indirect obstetric causes, more than 42 days but less than one year after termination of pregnancy
- **Severe pregnancy-related morbidity:**
  - accounts for a range of pregnancy-related complications that have serious short- and long-term consequences on an individual's health

Pregnancy-Related Morbidity

- Conditions such as hemorrhage, unplanned hysterectomy, eclampsia, stroke, sepsis, ARDS, cardiomyopathy, PE, depression, post-operative infection, to name a few.
- >70% of cases occur in the first 2 weeks after hospital discharge.
- Affects >50,000 people per year in the United States.
  - Nearly doubled in the last 10 years
  - Significantly higher rates of readmission in the 6 weeks postpartum --> stress on family/newborn, cost to healthcare system
- Who is most affected?
  - From 2012 to 2015, compared with White women, the incidence was:
  - **166% higher for Black women • 148% higher for American Indian/Alaska Native women • 122% higher for Hispanic women • 117% higher for Asian/Pacific Islander women**

Pregnancy-Related Morbidity

In Philadelphia…

16% of Philadelphia birthing people report symptoms of postpartum depression.
Asian and non-Hispanic Black birthing people report the highest rates of postpartum depression symptoms (20% and 17%, respectively), and non-Hispanic White people report the lowest rates of postpartum depression symptoms (12%).

In the US…

13% of birthing people report experiencing symptoms of postpartum depression.

Suicide is the manner of death in 7.4% of pregnancy-related deaths.

Severe Pregnancy-Related Morbidity
<20 | >35

50,000 – 60,000 people per year in the U.S.

178 people per year in **Philadelphia** (2016-2022)

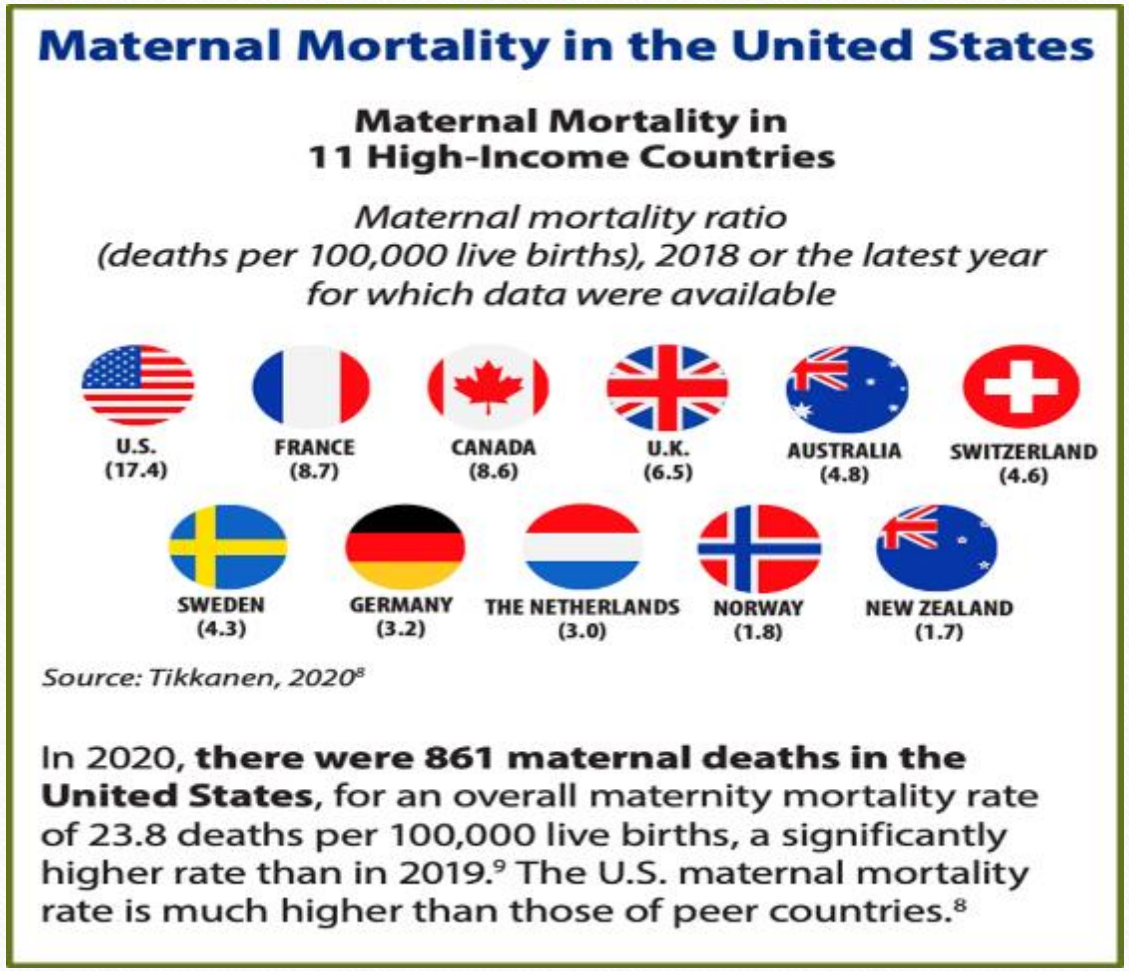
Severe Pregnancy-Related Morbidity rates in Philadelphia were **57% higher** than the national rate in 2011-2014.

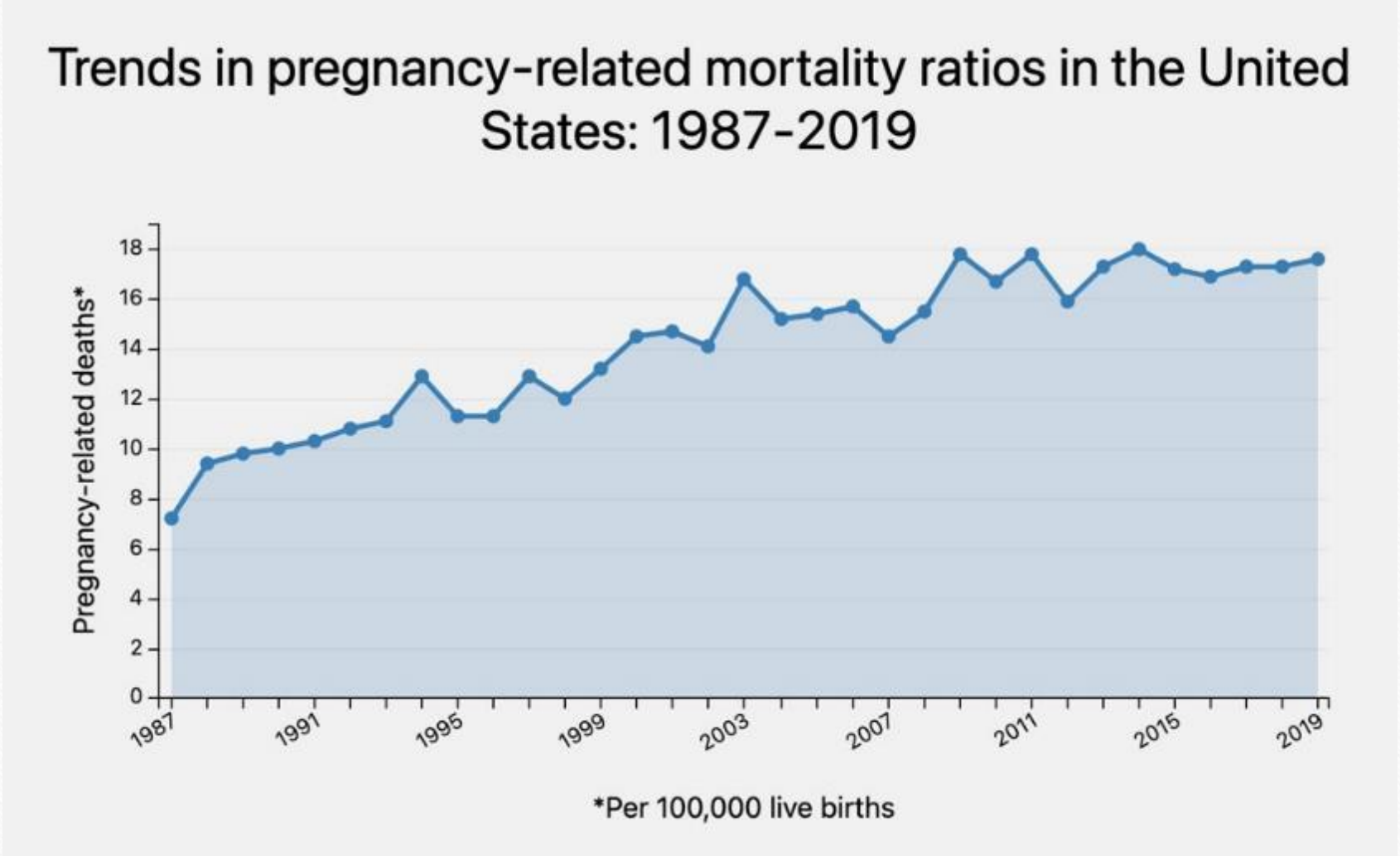
Severe Pregnancy-Related Morbidity Disease Index

- **Disseminated intravascular coagulation**: blood clotting disorder which causes uncontrollable bleeding and bruising
- **Renal Failure**: damage and disease to the kidneys
- **Infection and Sepsis**: critical contamination of the body by pathogens causing disease
- **Respiratory Distress**: difficulty breathing
- **Eclampsia**: pregnancy-related blood pressure disorder which causes seizures
- **Pulmonary Edema/Acute Heart Failure**: buildup of fluid in the lungs due to weakening and diminished function ofthe heart
- **Thrombotic Embolism**: blood clot in a vein which can travel to organs
- **Puerperal Cerebrovascular Disorder**: disease of blood vessels of the brain during pregnancy and the postpartumperiod (e.g. stroke)
- **Amniotic Fluid Embolism**: when the fluid that surrounds the fetus in utero travels into and affects the bloodstream
- **Hemorrhage**: loss of blood after childbirth

Despite high health care spending,

the U.S. has one of the highest pregnancy mortality rates in the industrialized world.





https://www.cdc.gov/reproductivehealth/maternal-mortality/pregnancy-mortality-surveillance-system.htm

Pregnancy-Related Mortality

About **1,205** individuals die each year in the United States from pregnancy-related causes (2021).

In **Philadelphia**, there are an average of 8 pregnancy-related deaths per year (2019-2023).

In Philadelphia

Between 2019 and 2023:

- **Most pregnancy-related deaths occurred in the postpartum period**, and **41% of deaths** occurred between 43 and 364 days after the end of pregnancy.
- Most pregnancy-related deaths were caused by **unintentional drug overdose**, accounting for 32% of pregnancy-related deaths.
- **Mental health conditions** (other than substance use disorder) were determined to be a contributing factor in 42% of pregnancy-related death cases.
- Black birthing people were **more likely** to die from a pregnancy-related cause compared to their White counterparts, irrespective of wealth or educational attainment;
- 90% of all pregnancy-related deaths were deemed to be **preventable** in this time period.

Unintentional drug overdose was the leading cause of pregnancy-related mortality in Philadelphia from 2019 to 2023, accounting for 32% of pregnancy-related deaths. Overdose deaths had been trending upward in Philadelphia and nationwide during this timeframe, particularly overdose deaths caused by opioids, and this trend is mirrored in maternal mortality data.

So Now What?!?

GOAL 1: Develop behavioral health systems to meet the specific needs of pregnant and postpartum people with Substance Use Disorder (SUD). Unintentional drug overdose is the leading cause of pregnancy-related mortality in Philadelphia.

GOAL 2: Improve infrastructure for mental health care and promote integration into maternal health care delivery systems. Perinatal mental health conditions are among the most common complications of pregnancy and postpartum.

GOAL 3: Optimize coordination and continuity of care Across health care and social service settings. Many maternal deaths occur outside of the health care setting and during transitions of care.

GOAL 4: Improve the understanding and recognition of pregnancy-related risks and complications across non-obstetric health care environments and the larger community. The need for heightened awareness about pregnancy-related risks and complications extends beyond obstetric care into broader health care environments and community settings.

GOAL 5: Implement strategies to establish medical trust with the community. Structural and historical harms have eroded community trust in health care.

Structural Solutions to Reduce Pregnancy-Related Morbidity

- Strengthen access to comprehensive health coverage across life course;
- Expand paid family and medical leave;
- Invest in economic security and family stability;
- Strengthen family-support policies within the criminal justice system;
- Improve access to childcare, early childhood supports, and nutrition programs;
- Integrate mental health across primary and reproductive health systems.

Why have these things happened?

- High pregnancy-related death rates in US
- Racial and ethnic disparities in pregnancy-related death

……these two discussions are very connected

The Uncomfortable Truth

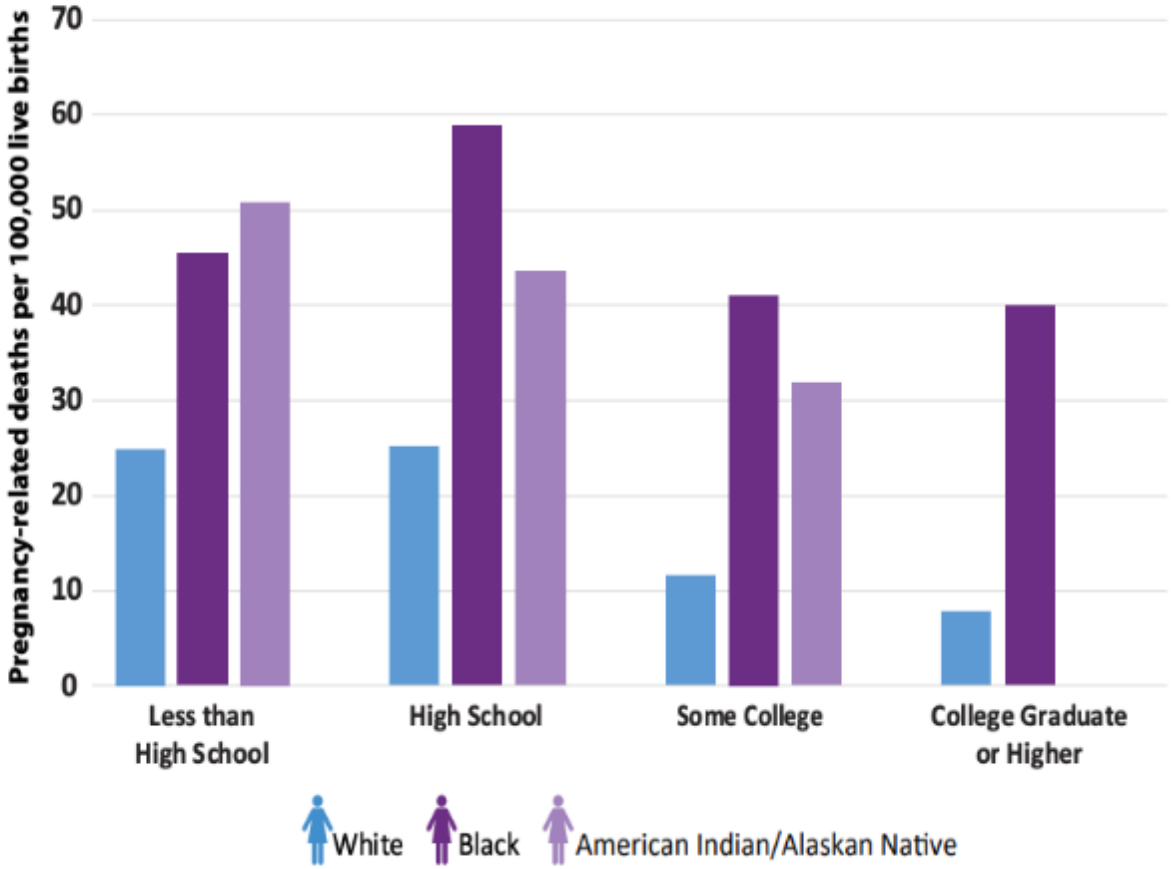
- INEQUITIES by race/ethnicity PERSIST even when known risk factors (e.g. smoking, obesity, etc) and confounding factors (e.g. educational attainment, income, etc) are taken into account.

- **Racism is our greatest public health crisis.**

Disparities by Education Level



The PRMR for black women with at least a college degree was 5 times as high as white women with a similar education.



https://www.cdc.gov/reproductivehealth/maternal-mortality/disparities-pregnancy-related-deaths/infographic.html

Why is the U.S. Pregnancy-Related Mortality Rate So High Compared to Other Industrialized Countries?



- Many barriers to accessing primary care providers in the US;
- Disparities in health care;
- Effects of chronic stress;
- Increasing prevalence of pre-existing health conditions;
  - Hypertension, type 2 diabetes and obesity have all seen significant increases, esp. since COVID
- Increase in incidence of mental health disorders/many barriers to care
  - Pre-pregnancy depression significantly increased
  - **At least** 1 in 8 pregnant people experience intrapartum or postpartum depression/anxiety (PMAD)
- Opioid Use, overdose, and suicide

Why is the U.S. Pregnancy-Related Mortality Rate So High Compared to Other Industrialized Countries?



- Overall shortage of pregnancy care workforce relative to number of births;
- Far fewer midwives for pregnant population;
- Decrease in profitability of obstetric care, with small hospitals and facilities closing L&D units nationwide;
- In 2019, 76.7% of pregnant people received early and adequate prenatal care;
- **People who do not receive prenatal care are 3-4 times more likely to die from pregnancy-related complications.**

Why is the U.S. Pregnancy-Related Mortality Rate So High Compared to Other Industrialized Countries?



- Lack of access to home visits, paid parental leave and ongoing primary healthcare compared to other countries
- Long-standing emphasis on infant health over health of the parent(s)
  - 6-8 weeks of postpartum care is the standard of care
- Lack of family and community support
  - Multigenerational effects of health care disparities

What do we do about this?

- Research
- Work on "safety bundles"
- Access to safe abortion
  - Abortion-restricted states have increased pregnancy-related death

    (*fewer providers, less access to care, disproportionately affecting patients with fewer resources*)
- Access to regular preventative care BEFORE pregnancy
- Access to prenatal care and regular health care in the year following pregnancy
- Individual and workforce/systems working towards health equity and antiracism

Research

Maternal Morbidity & Mortality Web Portal

The NIH Web Portal for Information about Maternal Morbidity and Mortality

Ongoing Studies

ORWH promotes ongoing studies that seek to advance the science on the health of women. This page includes ongoing studies focused on maternal health at both the NIH and broader U.S. Department of Health and Human Services.

- Chronic Hypertension and Pregnancy (CHAP) Project – This study evaluates the effectiveness and safety of a blood pressure treatment strategy during pregnancy to achieve targets that are recommended for nonpregnant reproductive-age adults (<140/90 mmHg) compared with the American College of Obstetricians and Gynecologists–recommended standard during pregnancy (no treatment unless blood pressure is severe).
- Final Research Plan: Aspirin Use to Prevent Morbidity and Mortality from Preeclampsia - Preventive Medication – This research plan is designed to study maternal health outcomes among women who are at increased risk of preeclampsia and use aspirin during pregnancy.
- nuMoM2b Heart Health Study – This study investigates the relationship between experiences during pregnancy and cardiovascular health 2 to 3½ years post-pregnancy.
- Human Placenta Project – This *Eunice Kennedy Shriver* National Institute of Child Health and Human Development–led project is a collaborative research effort to understand the role of the placenta in health and disease.
- Pregnancy Risk Assessment Monitoring System (PRAMS) – This Centers for Disease Control and Prevention (CDC) and health departments surveillance project collects jurisdiction-specific, population-based data on maternal attitudes and experiences before, during, and shortly after pregnancy. Researchers can use this data to investigate emerging issues in reproductive health.
- PregSource – This research project is designed to learn about pregnancy directly from pregnant women.
- Prevention of Iron Deficiency Anemia Post-delivery (PRIORITY) – This 2-arm, randomized-controlled trial study is focused on postpartum women who are diagnosed with moderate anemia based on a blood sample taken 6-48 hours after childbirth. The study will be implemented at eight sites in seven countries: Bangladesh, Democratic Republic of the Congo, Guatemala, India (Nagpur and Belagavi), Kenya, Pakistan, and Zambia.
- Researching COVID to Enhance Recovery (RECOVER) – This research initiative from the National Institutes of Health (NIH) seeks to understand, prevent, and treat post-acute sequelae (PASC), including Long COVID.

Safety Bundles

- AIM: The Alliance for Innovation on Maternal Health (AIM)
  - An ACOG initiative designed to lead in the development and implementation of patient safety bundles for the promotion of safe care for every U.S. birth



Distribution of Care

- **The right level of care:** regionalized systems of care can reduce pregnancy-related morbidity and mortality;
  - Distribution of providers and facilities
  - "The right resources to the right patient in the right place at the right time"
    - Stabilization and transfer to higher level of care within a system.
- **Increasing access to midwifery care.**
  - This has been less embraced in the US than in other developed countries.

Midwifery Model of Care

- Ensuring equitable care amongst all of our patients, regardless of race, primary language, creed, documentation status, and ability to pay.
- Ensuring that our care honors the autonomy, cultural context, and values of our patients via informed consent, shared decision-making, and understanding the sociopolitical contexts of our patient's lived experiences.
- Ensuring that we provide care that prioritizes **cultural humility** and sensitivity, including acknowledging the role of racism and actively working towards an antiracist healthcare system and society.
- Ensuring that our care comes from and reflects a **reproductive justice** lens.

Food for Thought

- How will this knowledge affect the care you give to individuals and families?
- How will your understanding of the social determinants of health and health disparities affect your advocacy for the care of your patients within your practice or the patient population served by your institution?
- In what ways might you consider the social conditions in which your patient grew up and now live as impacting their current health?

Resources

- https://orwh.od.nih.gov/mmm-portal
- https://www.acog.org/advocacy/policy-priorities/maternal-mortality-prevention
- https://saferbirth.org/patient-safety-bundles/
- https://www.gao.gov/assets/gao-23-105871.pdf
- https://www.kff.org/racial-equity-and-health-policy/issue-brief/racial-disparities-in-maternal-and-infant-health-current-status-and-efforts-to-address-them/



QUESTIONS?